

3. Sensitivity. It is the scale factor of a sensor that is measured in terms of change in output signal for every change in the input measured. It is measured in mV/gm.

4. Frequency response. It measures the limit of the frequency for the sensor-detected motion and the reported output. It is measured in hertz (Hz).

5. Sensitivity axis. The inputs detected by the accelerometers are always in reference to an axis. Single-axis accelerometers can detect inputs only along one plane whereas tri-axis accelerometers can detect inputs from any direction. So the tri-axis accelerometers are used in most of the applications.

6. Size and mass. Both the size and the mass of an accelerometer should be small as compared to that of the system to be monitored, otherwise it can affect and also change the characteristics of the object that is being tested.

The micro-electro-mechanical sensor (MEMS) accelerometer is special as compared to other accelerometers owing to its capability and small size. "The parameters to select an MEMS accelerometer are power consumption, sensitivity to 'g' force, ADC bit size, device calibration, axis support and presence of internal FIFO data buffer. Newer accelerometers support additional features such as free-fall detection, motion detection, gesture control and orientation detection algorithms, so designers can exploit these features depending on the application requirements," says Avinash Babu, senior project manager, Embedded Systems, Mistral Solutions.

So some points on the basis of which a gyroscope can be selected are:

Measurement range. It specifies the maximum angular speed measured by the gyro sensor. It is measured in degrees per second ($^{\circ}$ /sec).

Number of sensing axes. A gyroscope can measure angular rotation either in one, two or three axes; however, a multi-axis gyro has multiple single-axis gyros that are oriented orthogonally to one another.

Non-linearity. It specifies the closeness of the output voltage to linearity and is proportional to the actual

angular rate. It is measured either as an error in percentage or in parts per million (ppm).

Shock survivability. Since both the linear and angular rotations occur in a gyroscope, it is necessary to check the force it can withstand without falling. However, a gyroscope is expected to withstand very large shocks (measured in g's) without breaking.

Bandwidth. It is the number of measurements made in one second. It is quoted in hertz (Hz).

Angular random walk (ARW). It is the measurement of gyro noise in deg/hour^{1/2} or deg/sec^{1/2}.

Bias instability. It measures the goodness of a gyro in degrees per hour ($^{\circ}$ /hr).

Challenges faced by the design engineers

Technology brings challenges and the designers have to overcome all these in order to create something new. Babu says, "Placement of a sensor on the PCB is very crucial and is often overlooked. For optimal motion detection, a sensor needs to be placed away from the centre of the device, which helps to ensure better acceleration readings and makes them more significant in the detection of smaller motions from a higher moment of inertia than when they are placed right on the centre of the movement.

"Care must be taken to ensure that the package is not stressed by holes, or components on the PCB are not placed too close to the accelerometer. It is important to place the sensor where it is not vulnerable to be pushed or otherwise affected directly by the user's hands. It is good to avoid placing the sensor near components that may have large temperature variations, or that are constantly very hot, as this will affect the offset of the sensor."

"Footprint of the sensor, supply current and cost need to be minimal in all consumer applications. Functional safety and reliability of the sensor in vast operating conditions becomes a major challenge in automotive applications," explains Vikas Choudhary,

engineering manager, MEMS and Sensors at Analog Devices, Inc. The other challenges are power consumption, sensor integration and calibration.

"In battery-operated devices, the power consumed by the sensors should be minimised as much as possible and the sensors must support low-power mode in suspend, preferably with interrupt functions. This would help the host processor to get relieved from the continuously polling data. Even though enough care is taken during the layout phase, there are uninterrupted interferences to these sensors due to which they report inaccurate values. Thus there is a need for the calibration of these sensors and the use of a fine-tuned software for an optimum calibration," explains Babu.

This is not all. The other factor that can affect the working of these sensors to a larger extent is the electromagnetic interference, especially the very high frequency (VHF) EMF. T. Anand says, "Electromagnetic interference can produce false signal outputs. On the other hand, VHF-level electromagnetic interference can also cause intermodulation distortion and produce low-frequency measurement errors."

What is new in this segment?

Sony Ericsson's shake control and Samsung's motion play are examples of new technologies that make good use of an accelerometer. Microsoft uses accelerometer-based features in its Windows Embedded Compact for different touch-screen applications on Windows 8. Manufacturers also use them in devices to protect their hard drives from getting damaged in case of a free-fall.

Industries currently use MEMS-based accelerometers. They work on the principle of displacement of a small proof mass that is etched into the silicon surface of the IC and is suspended by small beams. As soon as the acceleration is applied to the device, a force is developed that displaces the mass. The support beams that act as a spring, and the fluid trapped in the IC that acts as a damper, result in the second-order lumped physical system which acts as